

26. Abstract Construction of Ideal Theory in the Algebraic Number Field.

J. Ber. d. DMV 33 (1924), p. 102

4. E. Noether, Göttingen: Abstract construction of ideal theory in the algebraic number field.

The abstract properties were stated that are completely equivalent to ideal theory in the algebraic number field, and that therefore characterize all rings which possess such an ideal theory. They are the rings with identity element and without zero divisors in which the double chain condition holds – every divisor chain of ideals breaks off in the finite, and likewise every multiple chain modulo every ideal different from the zero ideal – and which are integrally closed with respect to their quotient field. If the latter assumption is dropped, then theorems result concerning ideal theory in a finite order, some of which are already found without proof in Dedekind (3rd edition).

A detailed presentation will appear in the *Mathematische Annalen*.